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NAME: AmesHousing.txt
TYPE: Population
SIZE: 2930 observations, 82 variables
ARTICLE TITLE: Ames Iowa: Alternative to the Boston Housing Data Set

DESCRIPTIVE ABSTRACT: Data set contains information from the Ames Assessor’s Office used in
computing assessed values for individual residential properties sold in Ames, IA from 2006 to
2010.

SOURCES:
Ames, Iowa Assessor’s Office

VARIABLE DESCRIPTIONS:
Tab characters are used to separate variables in the data file. The data has 82 columns which
include 23 nominal, 23 ordinal, 14 discrete, and 20 continuous variables (and 2 additional
observation identifiers).

Order (Discrete): Observation number

PID (Nominal): Parcel identification number  - can be used with city web site for parcel review.

MS SubClass (Nominal): Identifies the type of dwelling involved in the sale.

       020      1-STORY 1946 & NEWER ALL STYLES
       030      1-STORY 1945 & OLDER
       040      1-STORY W/FINISHED ATTIC ALL AGES
       045      1-1/2 STORY - UNFINISHED ALL AGES
       050      1-1/2 STORY FINISHED ALL AGES
       060      2-STORY 1946 & NEWER
       070      2-STORY 1945 & OLDER
       075      2-1/2 STORY ALL AGES
       080      SPLIT OR MULTI-LEVEL
       085      SPLIT FOYER
       090      DUPLEX - ALL STYLES AND AGES
       120      1-STORY PUD (Planned Unit Development) - 1946 & NEWER
       150      1-1/2 STORY PUD - ALL AGES
       160      2-STORY PUD - 1946 & NEWER
       180      PUD - MULTILEVEL - INCL SPLIT LEV/FOYER
       190      2 FAMILY CONVERSION - ALL STYLES AND AGES

MS Zoning (Nominal): Identifies the general zoning classification of the sale.

       A        Agriculture
       C        Commercial
       FV       Floating Village Residential
       I        Industrial
       RH       Residential High Density
       RL       Residential Low Density
       RP       Residential Low Density Park
       RM       Residential Medium Density

Lot Frontage (Continuous): Linear feet of street connected to property

Lot Area (Continuous): Lot size in square feet

Street (Nominal): Type of road access to property

       Grvl     Gravel
       Pave     Paved

Alley (Nominal): Type of alley access to property

       Grvl     Gravel
       Pave     Paved
       NA       No alley access

Lot Shape (Ordinal): General shape of property

       Reg      Regular
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        IR1      Slightly irregular
        IR2      Moderately Irregular
        IR3      Irregular

  Land Contour (Nominal): Flatness of the property

        Lvl      Near Flat/Level
        Bnk      Banked - Quick and significant rise from street grade to building
        HLS      Hillside - Significant slope from side to side
        Low      Depression

  Utilities (Ordinal): Type of utilities available

        AllPub   All public Utilities (E,G,W,& S)
        NoSewr   Electricity, Gas, and Water (Septic Tank)
        NoSeWa   Electricity and Gas Only
        ELO      Electricity only

  Lot Config (Nominal): Lot configuration

        Inside   Inside lot
        Corner   Corner lot
        CulDSac  Cul-de-sac
        FR2      Frontage on 2 sides of property
        FR3      Frontage on 3 sides of property

  Land Slope (Ordinal): Slope of property

        Gtl      Gentle slope
        Mod      Moderate Slope
        Sev      Severe Slope

  Neighborhood (Nominal): Physical locations within Ames city limits (map available)

        Blmngtn  Bloomington Heights
        Blueste  Bluestem
        BrDale   Briardale
        BrkSide  Brookside
        ClearCr  Clear Creek
        CollgCr  College Creek
        Crawfor  Crawford
        Edwards  Edwards
        Gilbert  Gilbert
        Greens   Greens
        GrnHill  Green Hills
        IDOTRR   Iowa DOT and Rail Road
        Landmrk  Landmark
        MeadowV  Meadow Village
        Mitchel  Mitchell
        Names    North Ames
        NoRidge  Northridge
        NPkVill  Northpark Villa
        NridgHt  Northridge Heights
        NWAmes   Northwest Ames
        OldTown  Old Town
        SWISU    South & West of Iowa State University
        Sawyer   Sawyer
        SawyerW  Sawyer West
        Somerst  Somerset
        StoneBr  Stone Brook
        Timber   Timberland
        Veenker  Veenker

  Condition 1 (Nominal): Proximity to various conditions

        Artery   Adjacent to arterial street
        Feedr    Adjacent to feeder street
        Norm     Normal
        RRNn     Within 200' of North-South Railroad
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        RRAn     Adjacent to North-South Railroad
        PosN     Near positive off-site feature--park, greenbelt, etc.
        PosA     Adjacent to postive off-site feature
        RRNe     Within 200' of East-West Railroad
        RRAe     Adjacent to East-West Railroad

  Condition 2 (Nominal): Proximity to various conditions (if more than one is present)

        Artery   Adjacent to arterial street
        Feedr    Adjacent to feeder street
        Norm     Normal
        RRNn     Within 200' of North-South Railroad
        RRAn     Adjacent to North-South Railroad
        PosN     Near positive off-site feature--park, greenbelt, etc.
        PosA     Adjacent to postive off-site feature
        RRNe     Within 200' of East-West Railroad
        RRAe     Adjacent to East-West Railroad

  Bldg Type (Nominal): Type of dwelling

        1Fam     Single-family Detached
        2FmCon   Two-family Conversion; originally built as one-family dwelling
        Duplx    Duplex
        TwnhsE   Townhouse End Unit
        TwnhsI   Townhouse Inside Unit

  House Style (Nominal): Style of dwelling

        1Story   One story
        1.5Fin   One and one-half story: 2nd level finished
        1.5Unf   One and one-half story: 2nd level unfinished
        2Story   Two story
        2.5Fin   Two and one-half story: 2nd level finished
        2.5Unf   Two and one-half story: 2nd level unfinished
        SFoyer   Split Foyer
        SLvl     Split Level

  Overall Qual (Ordinal): Rates the overall material and finish of the house

        10       Very Excellent
        9        Excellent
        8        Very Good
        7        Good
        6        Above Average
        5        Average
        4        Below Average
        3        Fair
        2        Poor
        1        Very Poor

  Overall Cond (Ordinal): Rates the overall condition of the house

        10       Very Excellent
        9        Excellent
        8        Very Good
        7        Good
        6        Above Average
        5        Average
        4        Below Average
        3        Fair
        2        Poor
        1        Very Poor

  Year Built (Discrete): Original construction date

  Year Remod/Add (Discrete): Remodel date (same as construction date if no remodeling or additions)

  Roof Style (Nominal): Type of roof
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        Flat     Flat
        Gable    Gable
        Gambrel  Gabrel (Barn)
        Hip      Hip
        Mansard  Mansard
        Shed     Shed

  Roof Matl (Nominal): Roof material

        ClyTile  Clay or Tile
        CompShg  Standard (Composite) Shingle
        Membran  Membrane
        Metal    Metal
        Roll     Roll
        Tar&Grv  Gravel & Tar
        WdShake  Wood Shakes
        WdShngl  Wood Shingles

  Exterior 1 (Nominal): Exterior covering on house

        AsbShng  Asbestos Shingles
        AsphShn  Asphalt Shingles
        BrkComm  Brick Common
        BrkFace  Brick Face
        CBlock   Cinder Block
        CemntBd  Cement Board
        HdBoard  Hard Board
        ImStucc  Imitation Stucco
        MetalSd  Metal Siding
        Other    Other
        Plywood  Plywood
        PreCast  PreCast
        Stone    Stone
        Stucco   Stucco
        VinylSd  Vinyl Siding
        Wd Sdng  Wood Siding
        WdShing  Wood Shingles

  Exterior 2 (Nominal): Exterior covering on house (if more than one material)

        AsbShng  Asbestos Shingles
        AsphShn  Asphalt Shingles
        BrkComm  Brick Common
        BrkFace  Brick Face
        CBlock   Cinder Block
        CemntBd  Cement Board
        HdBoard  Hard Board
        ImStucc  Imitation Stucco
        MetalSd  Metal Siding
        Other    Other
        Plywood  Plywood
        PreCast  PreCast
        Stone    Stone
        Stucco   Stucco
        VinylSd  Vinyl Siding
        Wd Sdng  Wood Siding
        WdShing  Wood Shingles

  Mas Vnr Type (Nominal): Masonry veneer type

        BrkCmn   Brick Common
        BrkFace  Brick Face
        CBlock   Cinder Block
        None     None
        Stone    Stone

  Mas Vnr Area (Continuous): Masonry veneer area in square feet

  Exter Qual (Ordinal): Evaluates the quality of the material on the exterior
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        Ex       Excellent
        Gd       Good
        TA       Average/Typical
        Fa       Fair
        Po       Poor

  Exter Cond (Ordinal): Evaluates the present condition of the material on the exterior

        Ex       Excellent
        Gd       Good
        TA       Average/Typical
        Fa       Fair
        Po       Poor

  Foundation (Nominal): Type of foundation

        BrkTil   Brick & Tile
        CBlock   Cinder Block
        PConc    Poured Contrete
        Slab     Slab
        Stone    Stone
        Wood     Wood

  Bsmt Qual (Ordinal): Evaluates the height of the basement

        Ex       Excellent (100+ inches)
        Gd       Good (90-99 inches)
        TA       Typical (80-89 inches)
        Fa       Fair (70-79 inches)
        Po       Poor (<70 inches
        NA       No Basement

  Bsmt Cond (Ordinal): Evaluates the general condition of the basement

        Ex       Excellent
        Gd       Good
        TA       Typical - slight dampness allowed
        Fa       Fair - dampness or some cracking or settling
        Po       Poor - Severe cracking, settling, or wetness
        NA       No Basement

  Bsmt Exposure   (Ordinal): Refers to walkout or garden level walls

        Gd       Good Exposure
        Av       Average Exposure (split levels or foyers typically score average or above)
        Mn       Mimimum Exposure
        No       No Exposure
        NA       No Basement

  BsmtFin Type 1  (Ordinal): Rating of basement finished area

        GLQ      Good Living Quarters
        ALQ      Average Living Quarters
        BLQ      Below Average Living Quarters
        Rec      Average Rec Room
        LwQ      Low Quality
        Unf      Unfinshed
        NA       No Basement

  BsmtFin SF 1 (Continuous): Type 1 finished square feet

  BsmtFinType 2   (Ordinal): Rating of basement finished area (if multiple types)

        GLQ      Good Living Quarters
        ALQ      Average Living Quarters
        BLQ      Below Average Living Quarters
        Rec      Average Rec Room
        LwQ      Low Quality
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        Unf      Unfinshed
        NA       No Basement

  BsmtFin SF 2 (Continuous): Type 2 finished square feet

  Bsmt Unf SF (Continuous): Unfinished square feet of basement area

  Total Bsmt SF (Continuous): Total square feet of basement area

  Heating (Nominal): Type of heating

        Floor    Floor Furnace
        GasA     Gas forced warm air furnace
        GasW     Gas hot water or steam heat
        Grav     Gravity furnace
        OthW     Hot water or steam heat other than gas
        Wall     Wall furnace

  HeatingQC (Ordinal): Heating quality and condition

        Ex       Excellent
        Gd       Good
        TA       Average/Typical
        Fa       Fair
        Po       Poor

  Central Air (Nominal): Central air conditioning

        N        No
        Y        Yes

  Electrical (Ordinal): Electrical system

        SBrkr    Standard Circuit Breakers & Romex
        FuseA    Fuse Box over 60 AMP and all Romex wiring (Average)
        FuseF    60 AMP Fuse Box and mostly Romex wiring (Fair)
        FuseP    60 AMP Fuse Box and mostly knob & tube wiring (poor)
        Mix      Mixed

  1st Flr SF (Continuous): First Floor square feet

  2nd Flr SF (Continuous) : Second floor square feet

  Low Qual Fin SF (Continuous): Low quality finished square feet (all floors)

  Gr Liv Area (Continuous): Above grade (ground) living area square feet

  Bsmt Full Bath (Discrete): Basement full bathrooms

  Bsmt Half Bath (Discrete): Basement half bathrooms

  Full Bath (Discrete): Full bathrooms above grade

  Half Bath (Discrete): Half baths above grade

  Bedroom (Discrete): Bedrooms above grade (does NOT include basement bedrooms)

  Kitchen (Discrete): Kitchens above grade

  KitchenQual (Ordinal): Kitchen quality

        Ex       Excellent
        Gd       Good
        TA       Typical/Average
        Fa       Fair
        Po       Poor

  TotRmsAbvGrd    (Discrete): Total rooms above grade (does not include bathrooms)
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Functional (Ordinal): Home functionality (Assume typical unless deductions are warranted)

       Typ      Typical Functionality
       Min1     Minor Deductions 1
       Min2     Minor Deductions 2
       Mod      Moderate Deductions
       Maj1     Major Deductions 1
       Maj2     Major Deductions 2
       Sev      Severely Damaged
       Sal      Salvage only

Fireplaces (Discrete): Number of fireplaces

FireplaceQu (Ordinal): Fireplace quality

       Ex       Excellent - Exceptional Masonry Fireplace
       Gd       Good - Masonry Fireplace in main level
       TA       Average - Prefabricated Fireplace in main living area or Masonry Fireplace in
basement
       Fa       Fair - Prefabricated Fireplace in basement
       Po       Poor - Ben Franklin Stove
       NA       No Fireplace

Garage Type (Nominal): Garage location

       2Types   More than one type of garage
       Attchd   Attached to home
       Basment  Basement Garage
       BuiltIn  Built-In (Garage part of house - typically has room above garage)
       CarPort  Car Port
       Detchd   Detached from home
       NA       No Garage

Garage Yr Blt (Discrete): Year garage was built

Garage Finish (Ordinal) : Interior finish of the garage

       Fin      Finished
       RFn      Rough Finished
       Unf      Unfinished
       NA       No Garage

Garage Cars (Discrete): Size of garage in car capacity

Garage Area (Continuous): Size of garage in square feet

Garage Qual (Ordinal): Garage quality

       Ex       Excellent
       Gd       Good
       TA       Typical/Average
       Fa       Fair
       Po       Poor
       NA       No Garage

Garage Cond (Ordinal): Garage condition

       Ex       Excellent
       Gd       Good
       TA       Typical/Average
       Fa       Fair
       Po       Poor
       NA       No Garage

Paved Drive (Ordinal): Paved driveway

       Y        Paved
       P        Partial Pavement
       N        Dirt/Gravel
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Wood Deck SF (Continuous): Wood deck area in square feet

Open Porch SF (Continuous): Open porch area in square feet

Enclosed Porch (Continuous): Enclosed porch area in square feet

3-Ssn Porch (Continuous): Three season porch area in square feet

Screen Porch (Continuous): Screen porch area in square feet

Pool Area (Continuous): Pool area in square feet

Pool QC (Ordinal): Pool quality

       Ex       Excellent
       Gd       Good
       TA       Average/Typical
       Fa       Fair
       NA       No Pool

Fence (Ordinal): Fence quality

       GdPrv    Good Privacy
       MnPrv    Minimum Privacy
       GdWo     Good Wood
       MnWw     Minimum Wood/Wire
       NA       No Fence

Misc Feature (Nominal): Miscellaneous feature not covered in other categories

       Elev     Elevator
       Gar2     2nd Garage (if not described in garage section)
       Othr     Other
       Shed     Shed (over 100 SF)
       TenC     Tennis Court
       NA       None

Misc Val (Continuous): $Value of miscellaneous feature

Mo Sold (Discrete): Month Sold (MM)

Yr Sold (Discrete): Year Sold (YYYY)

Sale Type (Nominal): Type of sale

       WD       Warranty Deed - Conventional
       CWD      Warranty Deed - Cash
       VWD      Warranty Deed - VA Loan
       New      Home just constructed and sold
       COD      Court Officer Deed/Estate
       Con      Contract 15% Down payment regular terms
       ConLw    Contract Low Down payment and low interest
       ConLI    Contract Low Interest
       ConLD    Contract Low Down
       Oth      Other

Sale Condition (Nominal): Condition of sale

       Normal   Normal Sale
       Abnorml  Abnormal Sale -  trade, foreclosure, short sale
       AdjLand  Adjoining Land Purchase
       Alloca   Allocation - two linked properties with separate deeds, typically condo with a
garage unit
       Family   Sale between family members
       Partial  Home was not completed when last assessed (associated with New Homes)

SalePrice (Continuous): Sale price $$
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SPECIAL NOTES:
There are 5 observations that an instructor may wish to remove from the data set before giving it to students (a plot of SALE PRICE versus GR LIV AREA will indicate them quickly). Three of them are true outliers (Partial Sales that likely don't represent actual market values) and two of them are simply unusual sales (very large houses priced relatively appropriately). I would recommend removing any houses with more than 4000 square feet from the data set (which eliminates these 5 unusual observations) before assigning it to students.

STORY BEHIND THE DATA:
This data set was constructed for the purpose of an end of semester project for an undergraduate regression course. The original data (obtained directly from the Ames Assessor's Office) is used for tax assessment purposes but lends itself directly to the prediction of home selling prices. The type of information contained in the data is similar to what a typical home buyer would want to know before making a purchase and students should find most variables straightforward and understandable.

PEDAGOGICAL NOTES:
Instructors unfamiliar with multiple regression may wish to use this data set in conjunction with an earlier JSE paper that reviews most of the major issues found in regression modeling:

Kuiper , S. (2008), "Introduction to Multiple Regression: How Much Is Your Car Worth?", Journal of Statistics Education Volume 16, Number 3 (2008).

Outside of the general issues associated with multiple regression discussed in this article, this particular data set offers several opportunities to discuss how the purpose of a model might affect the type of modeling done. User of this data may also want to review another JSE article related directly to real estate pricing:

Pardoe , I. (2008), "Modeling home prices using realtor data", Journal of Statistics Education Volume 16, Number 2 (2008).

One issue is in regards to homoscedasticity and assumption violations. The graph included in the article appears to indicate heteroscedasticity with variation increasing with sale price and this problem is evident in many simple home pricing models that focus only on house and lot sizes. Though this violation can be alleviated by transforming the response variable (sale price), the resulting equation yields difficult to interpret fitted values (selling price in log or square root dollars). This situation gives the instructor the opportunity to talk about the costs (biased estimators, incorrect statistical tests, etc.) and benefits (ease of use) of not correcting this assumption violation. If the purpose in building the model is simply to allow a typical buyer or real estate agent to sit down and estimate the selling price of a house, such transformations may be unnecessary or inappropriate for the task at hand. This issue could also open into a discussion on the contrasts and comparisons between data mining, predictive models, and formal statistical inference.

A second issue closely related to the intended use of the model, is the handling of outliers and unusual observations. In general, I instruct my students to never throw away data points simply because they do not match a priori expectations (or other data points). I strongly make this point in the situation where data are being analyzed for research purposes that will be shared with a larger audience. Alternatively, if the purpose is to once again create a common use model to estimate a "typical" sale, it is in the modeler's best interest to remove any observations that do not seem typical (such as foreclosures or family sales).

REFERENCES:
Individual homes within the data set can be referenced directly from the Ames City Assessor webpage via the Parcel ID (PID) found in the data set. Note these are nominal values (non-numeric) so preceding 0's must be included in the data entry field on the website. Access to the database can be gained from the Ames site (http://www.cityofames.org/assessor/) by clicking on "property search" or by accessing the Beacon (http://beacon.schneidercorp.com/Default.aspx) website and inputting Iowa and Ames in the appropriate fields. A city map showing the location of all the neighborhoods is also available on the Ames site and can be accessed by clicking on "Maps" and then "Residential Assessment Neighborhoods (City of Ames Only)".

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